

PERSONAL INFORMATION

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EDUCATION BACKGROUND

- Ph.D. of Cyberspace Security 2015/09-2020/12
(Supervisor: Prof. **Hongwei Li: IEEE Fellow**)
School of Computer Science and Engineering
University of Electronic Science and Technology of China (UESTC)
- Visiting Ph.D. in Cyberspace Security 2019/08-2020/08
(Supervisor: Prof. **Robert H. Deng: IEEE Fellow**)
School of Information of Systems
Singapore Management University
- Bachelor of Information and Computing Science 2010/09-2014/06
School of Mathematical and Physical Science
Anhui Jianzhu University (AHJZU)

PROFESSIONAL EXPERIENCE

- Research Fellow 2021/03-2023/05
School of Computer Science and Engineering
Nanyang Technological University, Singapore
- Research Fellow 2023/05-2024/08
Department of Computer Science
City University of Hong Kong
- Professor 2024/08-present
School of Computer Science and Engineering
University of Electronic Science and Technology of China

RESEARCH INTERESTS

AI Security and Privacy, Data Security

AWARDS AND HONORS

- 2025 IEEE Transactions on Network and Service Management Exemplary Associate Editor Award, IEEE Communications Society
- 2025 IEEE 31st International Conference on Parallel and Distributed Systems (ICPADS) Best Presentation in Session Award, IEEE Computer Society
- 2025 IEEE Signal Processing Society Best Paper Award, IEEE Signal Processing Society
- 2025 R10 Young Professionals Outstanding Volunteer in Academic Award, IEEE Region 10 Young Professionals
- 2025 IEEE Systems, Man, and Cybernetics (SMC) Society's Technical Committee on Homeland Security Young Researcher Award, IEEE Systems, Man, and Cybernetics Society
- 2024 Computing's Top 30 Early Career Professionals, IEEE Computer Society
- 2024 IEEE Early Career Speaker, IEEE Computer Society
- 2024 Outstanding Youth Editor Award, Cybersecurity journal (Springer)
- 2023 IEEE 9th Intl Conference on Big Data Security on Cloud (BigDataSecurity) Best paper Award,

IEEE Computer Society

- 2022 Huawei Genius Young Talent Program, Huawei.
- 2021 Wu Wenjun First Prize of Artificial Intelligence Science and Technology Progress, Chinese Association for Artificial Intelligence
- 2020 IEEE 26th International Conference on Parallel and Distributed Systems (ICPADS) Best Paper Award, IEEE Computer Society

SELECTED PUBLICATIONS (CCF A papers: **82**; Google Citations: **7655**)

- [1] [TDSC 2026] Xinyuan Qian, Hongwei Li, **Guowen Xu*(Corresponding Author)**, Haoyong Wang, Hangcheng Cao, Cong Wu, Tianwei Zhang, Yuguang Fang. Perspicio: Safeguarding Against Non-consensual Photo Sharing over Social Networks. *IEEE Transactions on Dependable and Secure Computing*, 2026. **(CCF A)**
- [2] [TDSC 2026] Cong Wu, Jing Chen, Jian Shen, **Guowen Xu*(Corresponding Author)**, Yueming Wu, Haijun Wang, Hongwei Li, Yang Liu, Tao Xiang. Catching Scam Tokens with Temporal Graph Learning in Decentralized Finance. *IEEE Transactions on Dependable and Secure Computing*, 2026. **(CCF A)**
- [3] [TDSC 2026] Shuai Yuan, **Guowen Xu*(Corresponding Author)**, Hongwei Li, Rui Zhang, Xinyuan Qian, Hangcheng Cao, Qingchuan Zhao. FIGhost: Fluorescent Ink-based Stealthy and Flexible Backdoor Attacks on Physical Traffic Sign Recognition[J]. *IEEE Transactions on Dependable and Secure Computing*, 2026. **(CCF A)**
- [4] [TDSC 2026] Rui Zhang, Zihan Wang, Tianli Yang, Wenbo Jiang, Rui Zhang, Qingchuan Zhao, Hongwei Li, Yang Liu **Guowen Xu*(Corresponding Author)**. Hidden Tail: Adversarial Attack for Stealthy Resource Consumption against Vision-Language Models[J]. *IEEE Transactions on Dependable and Secure Computing*, 2026. **(CCF A)**
- [5] [TDSC 2026] Rui Zhang, Yun Shen, Hongwei Li, Wenbo Jiang, Hanxiao Chen, Yuan Zhang, **Guowen Xu*(Corresponding Author)**, Yang Zhang. Backdoor Complications: A Comprehensive Analysis and Mitigation of the Unforeseen Consequences of Backdoor Attacks[J]. *IEEE Transactions on Dependable and Secure Computing*, 2026. **(CCF A)**
- [6] [USENIX Security 2026] Jiayimei Wang, Tao Ni, **Guowen Xu**, Qingchuan Zha, Cong Wang. Adversarial Patch EXterminator: Zero-Shot and Patch-Agnostic Defense Framework Against Adversarial Patch Attacks[C]. *The 35th USENIX Security Symposium*, 2026 **(CCF A)**
- [7] [AAAI 2026] Xinyuan Qian, Haoyong wang, Hangcheng Cao, Shuai Yuan, Senkang Hu, Qingchuan Zhao, Hongwei Li, **Guowen Xu*(Corresponding Author)**. MartDE: A Privacy-Preserving and Cost-Efficient Evaluation Framework for Data Marketplaces[C]. *The Fortieth AAAI Conference on Artificial Intelligence*, 2025. **(CCF A)**
- [8] [AAAI 2026] Zihan Wang, Rui Zhang, Yu Liu, Wenshu Fan, Wenbo Jiang, Qingchuan Zhao, Hongwei Li, **Guowen Xu*(Corresponding Author)**. MPMA: Preference Manipulation Attack Against Model Context Protocol[C]. *The Fortieth AAAI Conference on Artificial Intelligence*, 2025. **(CCF A, Oral)**
- [9] [AAAI 2026] Zihan Wang, Rui Zhang, Hongwei Li, Wenshu Fan, Wenbo Jiang, Qingchuan Zhao, **Guowen Xu*(Corresponding Author)**. ConfGuard: A Simple and Effective Backdoor Detection for Large Language Models[C]. *The Fortieth AAAI Conference on Artificial Intelligence*, 2025. **(CCF A)**
- [10] [ACL 2026] Rui Zhang, Hongwei Li, Yun Shen, Xinyue Shen, Wenbo Jiang, **Guowen Xu*(Corresponding Author)**, Yang Liu, Michael Backes, Yang Zhang. The Art of (Mis)alignment:

- How Fine-Tuning Methods Effectively Misalign and Realign LLMs in Post-Training[C]. *The 64th Annual Meeting of the Association for Computational Linguistics*, 2026.(CCF A)
- [11] [AAAI 2026] Haonan An, Guang Hua, Hangcheng Cao, Zhengru Fang, Guowen Xu, Susanto Rahardja, Yuguang Fang. Removing Box-Free Watermarks for Image-to-Image Models via Query-Based Reverse Engineering[C]. *The Fortieth AAAI Conference on Artificial Intelligence*, 2025. (CCF A)
- [12] [TMC 2026] Senkang Hu, Yihang Tao, Guowen Xu, Xinyuan Qian, Yiqin Deng, Xianhao Chen, Sam Tak Wu Kwong, Yuguang Fang. CP-uniGuard: A Unified, Probability-Agnostic, and Adaptive Framework for Malicious Agent Detection and Defense in Multi-Agent Embodied Perception Systems[J]. *IEEE Transactions on Mobile Computing*, 2026. (CCF A)
- [13] [TIFS 2026] Jinghang Wen, Yongliang Chen, Ruoqin Tang, Xichen Yu, Guowen Xu, Lei Xue, Qingchuan Zhao, and Jian Weng. PriLabel: Towards Comprehensively Uncovering Omitted Disclosures in Privacy Labels of Android Apps on a Large Scale[J]. *IEEE Transactions on Information Forensics and Security*, 2025. (CCF A)
- [14] [TIFS 2026] Hanxiao Chen, Hongwei Li, Meng Hao, Jia Hu, Haomiao Yang, Tianwei Zhang, Guowen Xu. Sanitizer: Blazing-fast, Private and Robust Federated Learning. *IEEE Transactions on Information Forensics and Security*, 2026. (CCF A)
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- [16] [TDSC 2026] Shiqian Zhao, Meiqing Wu, Yi Xie, Tianlin Li, SiewKei Lam, Guowen Xu, Anran Li. Enhancing Stereo Matching Domain Generalization with Adversarial Domain Alignment. *IEEE Transactions on Dependable and Secure Computing*, 2026. (CCF A)
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- [21] [TDSC 2025] Jianfei Sun, Guowen Xu*(Corresponding Author), Hongwei Li, Tianwei Zhang, Cong Wu, Xuehuan Yang, Robert H. Deng. Sanitizable Cross-domain Access Control with Policy-driven Dynamic Authorization[J]. *IEEE Transactions on Dependable and Secure Computing*, 2025. (CCF A)

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- [23] [NeurIPS 2025] Shuai Yuan, Xingshuo Han, Hongwei Li, **Guowen Xu*(Corresponding Author)**, Wenbo Jiang, Tao Ni, Qingchuan Zhao, Yuguang Fang. The Fluorescent Veil: A Stealthy and Effective Physical Adversarial Patch Against Traffic Sign Recognition[C]. *The Thirty-Ninth Annual Conference on Neural Information Processing Systems*, 2025. (CCF A)
- [24] [ICML 2025] Rui Zhang, Yun Shen, Hongwei Li, Wenbo Jiang, Hanxiao Chen, Yuan Zhang, **Guowen Xu*(Corresponding Author)**, Yang Zhang. The Ripple Effect: On Unforeseen Complications of Backdoor Attacks[C]. *International Conference on Machine Learning*, 2025. (CCF A)
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- [32] [TIFS 2025] Hangcheng Cao, **Guowen Xu**, Ziyang He, Xuan Zhang, Shengmin Xu, Xinyuan Qian, Anjia Yang, Jianting Ning. The Lives of Others: Snooping on Smartphone Usage Behaviors via Attention-enabled Multi-channel Spatiotemporal Information Fusion[J]. *IEEE Transactions on Information Forensics and Security (TIFS)*, 2025. (CCF A)
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PROJECT EXPERIENCES

- [1] [2027.01-2029.12] Research on Key Technologies of Full-Link Security Evaluation and Protection for Intelligent Agents, Young Teachers Fund of the Fok Ying Tung Education Foundation, **PI**, **¥180K**
- [2] [2026.01-2028.12] Research on Security-Critical Technologies for Autonomous Driving Based on Multimodal Large Models, National Science Fund Program for Young Scientists of China, **PI**, Amount: **¥300K**.
- [3] [2026.01-2029.12] XXX Platform Design and Implementation. Key R&D Project of XXX. **PI**, Amount: **¥9.6 million**.
- [4] [2026.01-2027.12] Research on Key Technologies for Integrated Attack-Defense of Multimodal Large Models for Autonomous Driving. General Program of the Sichuan Provincial Natural Science Foundation. **PI**, Amount: **¥150K**.
- [5] [2025.01-2027.12] Data Security, National Science Fund Program for Excellent Young Scientists (Overseas) of China, **PI**, Amount: **¥2 million**.
- [6] [2024.09-2027.09] Key Security and Privacy Issues on Machine Learning, **PI**, Start-up at UESTC, Amount: **¥2 million**.

ACADEMIC ACTIVITIES

➤ Editorial Board:

- IEEE Transactions on Pattern Analysis and Machine Intelligence, 2025-present
- IEEE Transactions on Dependable and Secure Computing, 2025-present
- IEEE Transactions on Information Forensics and Security, 2024-present
- ACM Computing Surveys, 2026-present
- IEEE Transactions on Consumer Electronics, 2026-present
- IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2025-present
- IEEE Transactions on Circuits and Systems for Video Technology, 2024-2026
- IEEE Transactions on Network and Service Management, 2023-2026
- IEEE Open Journal of Signal Processing (OJSP), 2023-2026
- ACM Digital Threats: Research and Practice, 2023-2026
- Pattern Recognition (Elsevier), 2024-2026
- Information Fusion (Elsevier), 2024-2026
- Cybersecurity (Springer), 2024-2026
- Lead Guest Editor of ACM Transactions on Autonomous and Adaptive Systems (TAAS), Special Issue on Trustworthy Security and Privacy-AI Powered Autonomous Driving, 2024-2025

➤ Technical Committee:

- 2027 **Area Chair** of AAAI Conference Artificial Intelligence (AAAI)
- 2027 **Area Chair** of International Joint Conference on Neural Networks (IJCNN)

- 2026 **Area Chair** of Neural Information Processing Systems (NeurIPS)
- 2026-2028 Member of IEEE Region 10 Young Professionals Committee (YPC)
- 2026 **Associate Chair** ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW)
- 2026 **Associate Chair** of ACM CHI conference on Human Factors in Computing Systems (CHI)
- 2026 **Area Chair** of International Conference on Machine Learning (ICML)
- 2026 **Area Chair** of IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)
- 2026 **Area Chair** of International Conference on Learning Representations (ICLR)
- 2026 **Area Chair** of ACM Conference on Knowledge Discovery and Data Mining (KDD)
- 2026 **Senior Program Committee** of AAAI Conference Artificial Intelligence (AAAI)
- 2026 **Area Chair** of International Joint Conference on Neural Networks (IJCNN)
- 2026 The ACM Conference on Computer and Communications Security (CCS)
- 2026 The 35th USENIX Security Symposium (USENIX Security)
- 2025 **Area Chair** of ACM Conference on Knowledge Discovery and Data Mining (KDD)
- 2025 **Area Chair** of International Conference on Machine Learning (ICML)
- 2025 **Associate Chair** of 28th ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW)
- 2025 **Area Chair** of International Conference on Learning Representations (ICLR)
- 2025 **Senior Program Committee** of AAAI Conference Artificial Intelligence (AAAI)
- 2025 **Area Chair** of International Joint Conference on Neural Networks (IJCNN)
- 2025 The 35-th IEEE Visualization and Visual Analytics Conference (VIS)
- 2025 The 37th International Conference on Computer Aided Verification (CAV)
- 2025 The 32nd IEEE Conference on Virtual Reality (VR)
- 2025 International Conference on Autonomous Agents&Multiagent Systems (AAMAS)
- 2025 The 30th Annual ACM Conference on Intelligent User Interfaces (IUI)
- 2025 The 29th Financial Cryptography and Data Security Conference (FC)
- 2024 **Area Chair** of International Conference on Machine Learning (ICML)
- 2024 Thirty-eighth Conference on Neural Information Processing Systems (NeurIPS)
- 2024 Annual Computer Security Applications Conference (ACSAC)
- 2023 The ACM Web Conference
- 2023-2024 Thirty-Seventh AAAI Conference on Artificial Intelligence (AAAI)
- 2022 Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS)

➤ **Others:**

- Technical Review Board of IEEE Transactions on Parallel and Distributed Systems (TPDS), 2025-present
- Distinguished Reviewers Board of ACM Transactions on the Web, 2022-present
- Mentor of IEEE Signal Processing Mentorship Academy (SigMA), 2025-present